Appendix I. Honduras: Debt Sustainability Analysis

Under the policies envisaged by the proposed program, the debt sustainability analysis (DSA) shows improved public debt sustainability prospects compared to the previous DSA. However, given risks to the macroeconomic and fiscal outlook, Honduras' risk of external debt distress is assessed to remain moderate. External public debt indicators, except for one, are expected to remain below their indicative thresholds. Against this background, an improved rating of low risk of debt distress could be considered in a future DSA provided that the program's fiscal consolidation and growth targets are achieved.

A. Background

Honduras' public debt ratios continued to increase in 2013, reflecting weak fiscal policies. The fiscal position deteriorated in recent years, with the overall deficit of the combined public sector (CPS) rising to 7½ percent of GDP in 2013, from 2.8 percent in 2011.[1] By end 2013, the public debt ratio reached 45 percent of GDP. While the share of commercial external debt rose in 2013, following the placement of US$1 billion (5.4 percent of GDP) in global bonds,[2] about 80 percent of public external debt is to official creditors. A substantial portion of external public debt has been contracted on concessional terms, with a grant element averaging about 20 percent in 2013 (Table A2).

Public debt is mostly with foreign creditors. While the share of domestic debt has risen in recent years, over 60 percent of total public debt is accounted for by external creditors. The main external lenders to Honduras are the Inter-American Development Bank (IADB), the Central American Bank for Economic Integration, and the World Bank. Debt to these institutions carries long maturities, although given current lending and discount rates, only the IADB continues to provide concessional loans (i.e., with a grant element of at least 35 percent) to Honduras. Domestic public debt is mainly to commercial banks, has a shorter—though rising— maturity (about 3 years), and carries a higher real interest rate (Tables A1 and A2).

B. Underlying Assumptions

Economic growth. The baseline scenario assumes that growth in Honduras will rise to 3 percent in 2014–15 and strengthen further in subsequent years, before stabilizing at about 3¾ percent—broadly in line with its estimated potential. This reflects an improvement from the previous DSA, where growth was expected to remain at about 3 percent in the medium term. However, growth is

[1] The combined public sector includes the central government, local governments, decentralized agencies, social security institutions, the central bank, and public enterprises. The DSA baseline includes costs from PPP-related liabilities.

[2] The US$1 billion bond was placed in two tranches. The first tranche (US$0.5 billion, 11 years, 7.5 percent interest rate) was placed in March 2013; and the second tranche (US$0.5 billion, 7 years, at 8.75 percent) was placed in December 2013.

still projected to remain below the historical average (4 percent) throughout the DSA period. The acceleration of growth over the medium term is predicated on more favorable external conditions (notably better terms of trade and a growth pick up in the US) and the recovery of the coffee sector from leaf-rust disease, as well as improved confidence and higher private investment from stronger economic policy management. These factors are expected to offset the dampening effect of fiscal consolidation on growth, which in Honduras is expected to be modest given the high degree of openness of the economy.

Fiscal policy. The baseline scenario incorporates a cumulative fiscal adjustment of about 6½ percent of GDP during the program period (2014–17), with the primary deficit of the CPS declining to 1¾ percent of GDP and ½ percent of GDP by 2017, respectively (a significant improvement from the previous DSA, where the overall balance was expected to reach about 4¼ percent of GDP in 2017). The overall CPS deficit continues to decline slowly over the medium term as debt and interest payments start to fall, stabilizing at about 1 percent of GDP over the longer term (consistent with a primary deficit of ¼ percent of GDP).[3] The fiscal adjustment features a combination of higher tax revenues (about 2 percent of GDP) and lower expenditure (about 3 percent of GDP), mostly at the central government level. It also reflects a substantial improvement in the finances of the state-owned electricity company (ENEE)—whose deficit (1.8 percent of GDP in 2013) is projected to be eliminated by 2018. This improvement comes from higher tariffs, reduction of losses, and wage bill savings.

Public sector financing. Financing for the public sector is assumed to come mainly from external sources, especially multilateral lenders. It is, however, assumed that the financing mix changes toward less concessional sources compared to recent years (Table A2), including within multilateral institutions. The baseline scenario assumes the placement of two additional global bonds in 2015–16, totaling about US$300 million (1.4 percent of GDP), to help meet government financing needs. These bonds are assumed to carry an interest rate of 7 percent (significantly above current yields in the secondary market, which have hovered around 6 percent in recent weeks) and 10-year maturity. All global bonds are assumed to be refinanced through the issuance of new, similar bonds. Other financing needs are met through official, mostly multilateral, financing. The DSA includes existing domestic arrears in the stock of public debt, and assumes that arrears are paid down with domestic bonds bearing market rates in 2015.

External sector. The external current account deficit is projected to decline from 9½ percent in 2013 to 6.4 percent by 2017, and stabilize at about 5¼ percent of GDP over the longer term. The previous DSA projected a weaker external position— with the current account deficit stabilizing at about 6½ percent of GDP over the medium term—reflecting mainly a significantly smaller fiscal consolidation effort. The improvement in the current account during the program period reflects more favorable external conditions (including continued recovery of worker remittances), lower

[3] The primary deficit differs from that reported in Table A1. The latter reflects the primary deficit calculation in the DSA framework, which does not take into account interest revenue.

income payments, and a stronger macroeconomic policy mix (fiscal consolidation and a weaker real exchange rate). As in the previous DSA, the current account deficit continues to be financed through foreign direct investment and, to a lesser extent, public sector borrowing, with very small private sector flows. This allows international reserves to increase to about 4 months of imports by 2017 and settle at close to 4½ months of imports over the longer term.

C. Public and External Debt Sustainability Analysis

Public debt ratios are expected to peak in 2017, and start to decline subsequently (Table A1)**.** Public debt is projected to peak at about 51 percent of GDP in 2017 (up from 45 percent of GDP in 2013) and start falling slowly as fiscal consolidation proceeds and interest payments decline, reaching 45 percent of GDP by 2024. This represents a significant improvement from the previous DSA, which projected a public debt-to-GDP ratio of 67 percent by 2024. In present value terms, the public-debt ratio is expected to peak at 46 percent of GDP and fall to about 40 percent of GDP in the early 2020s. The public debt dynamics remains somewhat vulnerable to both policy-related and exogenous shocks, especially to those related to economic growth and fiscal policy (Table A3).

The evolution of external debt ratios is projected to follow a similar pattern (Table A2). External debt ratios are projected to peak at 41 percent of GDP in 2017, from about 36 percent of GDP in 2013. The share of external debt in total public debt rises slowly to about two-thirds by the mid-2020s and stabilizes at that level. The ratios for the PV of public debt and public debt service remain below their indicative thresholds in the baseline scenario throughout the projection period.[4,5] (Table A4). No alternative scenario leads to breaching the thresholds for the PV of debt. The threshold for the ratio of debt service to revenue is exceeded in two separate years (2020 and 2025) under the most extreme shock, with the spikes reflecting the amortization of global bonds. The ratio of debt service to exports remains well below its threshold under all scenarios.

A customized scenario with several negative shocks shows that external public debt indicators remain below their indicative thresholds, except for the debt service to revenue ratio. The scenario aims at capturing key risks and vulnerabilities in Honduras. It includes negative shocks to GDP growth, which could arise from weaker-trading partner growth ; harder external financing conditions that could be associated with negative developments in international financial markets; a weaker fiscal position from possible slippages in the implementation of adjustment measures; and the assumption of PPP-related contingent liabilities by the government.[6] In this scenario, external

[4] Debt threshold for Honduras are based on the average ratio in the Country Policy and Institutional Assessment (CPIA) for 2011-13, which implies a medium policy performance rating.

[5] Remittances are included in the external DSA (Honduras meets the criteria for its inclusion). In the last three years, they were around 16 percent of GDP—still below pre-global crisis levels but expected to continue to recover over the next few years—and close to 40 percent of exports of goods and services.

[6] In the customized scenario, (i) real GDP growth is 2.5 percent from 2014 to 2017, and rises 0.2 percentage points per year to reach 3.3 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, with the rate of interest on Eurobonds rising to 8.5 percent, starting in 2015; (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the

(continued)

public debt rises initially but remains below the indicative threshold and starts to decline around the mid-2020s. The ratio of debt service to exports also remains below its indicative threshold. The threshold for the ratio of debt service to revenue is exceeded once (in 2020, owing to the global bond repayment, as mentioned above) and approaches the threshold on two subsequent dates.

D. Conclusion

Public and external debt indicators have strengthened compared to previous DSA, but the risk of debt distress remains moderate. The improvement in debt indicator reflects the stronger macroeconomic policies that would underpin the Fund-supported program, which would translate into substantially lower fiscal and current account deficits. It is also aided by somewhat higher economic growth compared to the previous DSA. However, given risks to the macroeconomic and fiscal outlook, Honduras' risk of external debt distress is assessed to remain moderate. External public debt indicators are expected to stay below their indicative thresholds (except for the ratio of debt service to revenue), provided that the program's fiscal consolidation and growth targets are achieved. In that context, a rating of low risk of debt distress could be considered in a future DSA.

baseline (this increase fades out partially at a constant rate of 15 percent during the DSA period, e.g., compared to the baseline, the deficit is higher by 1.7, 1.0 and 0.5 percent of GDP respectively in 2016, 2019 and 2024; and (iv) contingent liabilities (estimated by the authorities at about 2 percent of GDP) related to existing PPP projects are assumed to materialize over the next three years and result in an increase in public external debt.

Table A1. Honduras: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	32.1	34.4	45.1			47.8	49.4	50.5	50.8	50.5	49.9		45.1	36.1	
of which: foreign-currency denominated	18.1	19.7	28.1			29.7	31.8	32.9	33.1	33.0	32.6		30.0	23.4	
Change in public sector debt	2.3	2.3	10.8			2.6	1.6	1.1	0.3	-0.3	-0.6		-0.8	-0.7	
Identified debt-creating flows	-0.3	2.7	6.8			3.4	0.8	0.0	-0.6	-1.0	-1.2		-1.6	-1.0	
Primary deficit 2/	1.7	3.0	5.7	2.3	1.6	3.5	0.5	-0.8	-1.2	-1.5	-1.7	-0.2	-1.5	-1.0	-1.3
Revenue and grants	23.1	22.5	22.9			24.5	24.5	25.0	25.2	25.2	25.2		25.2	25.2	
of which: grants	1.1	1.0	0.8			1.1	0.8	0.6	0.5	0.4	0.4		0.4	0.4	
Primary (noninterest) expenditure	24.9	25.5	28.6			28.0	25.0	24.1	23.9	23.7	23.5		23.8	24.3	
Automatic debt dynamics	-2.1	-0.3	1.1			-0.1	0.3	0.8	0.6	0.5	0.5		-0.2	-0.1	
Contribution from interest rate/growth differential	-1.2	-0.8	0.5			-0.2	0.0	0.1	-0.1	-0.3	-0.3		-0.3	-0.2	
of which: contribution from average real interest rate	-0.1	0.4	1.4			1.1	1.3	1.7	1.6	1.5	1.5		1.3	1.2	
of which: contribution from real GDP growth	-1.1	-1.3	-0.9			-1.3	-1.4	-1.6	-1.7	-1.8	-1.8		-1.6	-1.3	
Contribution from real exchange rate depreciation	-0.8	0.5	0.6			0.1	0.4	0.7	0.7	0.8	0.8		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.6	-0.4	4.0			-0.8	0.8	1.1	0.9	0.7	0.7		0.8	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	39.9			42.6	44.2	45.4	45.6	45.2	44.6		40.8	32.4	
of which: foreign-currency denominated	...	...	22.8			24.5	26.7	27.7	27.9	27.8	27.3		25.7	19.7	
of which: external	...	...	22.8			24.5	26.7	27.7	27.9	27.8	27.3		25.7	19.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	5.7	6.8	10.1			9.3	7.4	5.2	3.0	3.0	2.8		2.9	2.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	174.4			174.0	180.6	181.6	181.0	179.2	176.7		161.8	128	
PV of public sector debt-to-revenue ratio (in percent)	...	...	180.5			181.9	186.6	186.0	184.8	182.3	179.8		164.6	131	
of which: external 4/	...	...	103.2			104.7	112.5	113.6	113.2	112.0	110.1		103.6	79.4	
Debt service-to-revenue and grants ratio (in percent) 5/	17.1	17.0	19.2			23.7	28.3	24.0	16.9	17.8	17.9		17.4	14.3	
Debt service-to-revenue ratio (in percent) 5/	17.9	17.8	19.9			24.7	29.2	24.6	17.3	18.1	18.2		17.7	14.5	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.5	0.7	-5.1			0.9	-1.2	-2.0	-1.5	-1.2	-1.1		-0.7	-0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.8	4.1	2.6	4.1	2.7	3.0	3.0	3.3	3.5	3.7	3.7	3.4	3.7	3.7	3.7
Average real interest rate on total debt (in percent)	-3.6	0.5	4.2			0.4	0.7	1.9	1.7	1.6	1.6	1.3	1.7	1.9	1.9
Average nominal interest rate on forex debt (in percent)	1.7	1.8	2.0	1.7	0.2	3.6	3.6	3.7	3.8	3.8	3.8	3.7	3.8	4.1	4.0
Average real interest rate on domestic debt (in percent)	-0.5	3.3	9.0	1.4	3.9	3.6	4.9	6.7	6.2	5.9	5.8	5.5	5.6	5.5	5.6
Real exchange rate depreciation (in percent, + indicates depreciation)	-4.9	3.0	3.2	-2.1	3.4	0.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.8	3.6	1.5	5.9	2.1	5.3	5.7	4.6	4.8	4.7	4.7	5.0	4.2	4.2	4.2
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.1	6.7	15.1	5.4	6.8	0.9	-8.1	-0.1	2.7	2.6	3.0	0.2	4.1	3.7	3.9
Grant element of new external borrowing (in percent)	...	...	...	...	...	13.0	8.5	7.5	13.6	14.0	16.2	12.2	6.5	8.1	...

Sources: National authorities and Fund staff estimates and projections.

1/ Combined public sector.

2/ Primary deficit does not take into account interest earnings.

3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

4/ Revenues excluding grants.

5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A2. Honduras: External Debt Sustainability Framework, Baseline Scenario, 2011–2034 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 8/ Average	Standard 8/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**23.8**	**26.2**	**35.9**			**37.8**	**39.9**	**40.9**	**41.1**	**40.9**	**40.4**		**37.6**	**30.7**	
of which: public and publicly guaranteed (PPG)	18.1	19.7	28.1			29.7	31.8	32.9	33.1	33.0	32.6		30.0	23.4	
Change in external debt	-0.2	2.4	9.7			1.9	2.1	1.0	0.2	-0.2	-0.5		-0.8	-0.6	
Identified net debt-creating flows	-0.4	2.8	4.2			1.8	1.0	0.6	0.1	-0.2	-0.6		-0.8	-0.8	
Non-interest current account deficit	**7.5**	**8.0**	**8.8**	**6.8**	**3.7**	**6.6**	**5.7**	**5.3**	**4.9**	**4.5**	**4.1**		**3.9**	**3.9**	3.9
Deficit in balance of goods and services	20.2	19.5	20.6			19.6	19.3	19.1	18.8	18.9	18.9		18.9	18.9	
Exports	51.0	50.9	47.8			47.6	47.8	48.1	48.5	48.9	49.2		49.2	49.2	
Imports	71.2	70.3	68.4			67.2	67.1	67.2	67.3	67.8	68.1		68.1	68.1	
Net current transfers (negative = inflow)	-17.8	-17.8	-18.4	-19.0	2.4	-18.4	-18.6	-18.7	-19.0	-19.5	-19.9		-20.1	-20.0	-20.1
of which: official	-1.3	-1.2	-0.8			-0.8	-1.0	-0.9	-0.9	-0.9	-0.9		-0.9	-0.9	
Other current account flows (negative = net inflow)	5.0	6.3	6.6			5.4	5.1	5.0	5.0	5.1	5.1		5.0	5.0	
Net FDI (negative = inflow)	**-5.7**	**-4.6**	**-5.4**	**-5.9**	**1.2**	**-5.0**	**-5.0**	**-4.9**	**-4.9**	**-4.8**	**-4.8**		**-4.8**	**-4.8**	-4.8
Endogenous debt dynamics 2/	**-2.1**	**-0.6**	**0.7**			**0.2**	**0.2**	**0.2**	**0.1**	**0.1**	**0.1**		**0.1**	**0.1**	
Contribution from nominal interest rate	0.5	0.5	0.7			1.2	1.3	1.4	1.5	1.5	1.5		1.4	1.2	
Contribution from real GDP growth	-0.8	-0.9	-0.7			-1.0	-1.1	-1.3	-1.4	-1.5	-1.5		-1.3	-1.1	
Contribution from price and exchange rate changes	-1.8	-0.2	0.7			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**0.1**	**-0.4**	**5.5**			**0.1**	**1.1**	**0.3**	**0.1**	**0.1**	**0.2**		**0.1**	**0.2**	
of which: exceptional financing	0.0	0.0	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	30.7			32.7	34.7	35.7	35.8	35.6	35.1		33.3	27.0	
In percent of exports	...	...	64.1			68.6	72.6	74.2	73.8	72.9	71.4		67.7	54.9	
PV of PPG external debt	**...**	**...**	**22.8**			**24.5**	**26.7**	**27.7**	**27.9**	**27.8**	**27.3**		**25.7**	**19.7**	
In percent of exports	**...**	**...**	**47.7**			**51.5**	**55.8**	**57.6**	**57.5**	**56.8**	**55.5**		**52.3**	**40.1**	
In percent of government revenues	**...**	**...**	**103.2**			**104.7**	**112.5**	**113.6**	**113.2**	**112.0**	**110.1**		**103.6**	**79.4**	
Debt service-to-exports ratio (in percent)	**8.3**	**15.8**	**16.9**			**15.9**	**16.8**	**16.9**	**17.4**	**17.5**	**17.7**		**17.9**	**16.3**	
PPG debt service-to-exports ratio (in percent)	**1.9**	**1.4**	**1.6**			**3.3**	**3.6**	**3.9**	**4.6**	**4.9**	**5.3**		**6.0**	**4.9**	
PPG debt service-to-revenue ratio (in percent)	**4.5**	**3.4**	**3.5**			**6.7**	**7.3**	**7.6**	**9.0**	**9.7**	**10.4**		**11.9**	**9.7**	
Total gross financing need (Billions of U.S. dollars)	1.4	2.4	2.4			2.1	2.1	2.1	2.2	2.2	2.2		2.6	4.1	
Non-interest current account deficit that stabilizes debt ratio	7.7	5.6	-0.9			4.7	3.7	4.4	4.7	4.7	4.6		4.6	4.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.8	4.1	2.6	4.1	2.7	3.0	3.0	3.3	3.5	3.7	3.7	3.4	3.7	3.7	3.7
GDP deflator in US dollar terms (change in percent)	8.1	0.7	-2.7	4.3	3.6	2.2	0.6	-0.4	-0.2	-0.3	-0.3	0.3	1.4	1.6	1.2
Effective interest rate (percent) 5/	2.3	2.3	2.7	2.4	0.3	3.7	3.7	3.7	3.8	3.8	3.8	3.8	3.9	4.0	3.9
Growth of exports of G&S (US dollar terms, in percent)	24.3	4.6	-6.1	8.2	13.5	4.7	4.1	3.6	4.1	4.2	3.9	4.1	5.1	5.4	4.9
Growth of imports of G&S (US dollar terms, in percent)	24.8	3.5	-2.9	7.8	18.9	3.4	3.5	3.1	3.5	4.1	3.8	3.6	5.1	5.4	4.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	13.0	8.5	7.5	13.6	14.0	16.2	12.2	6.5	8.1	8.2
Government revenues (excluding grants, in percent of GDP)	22.0	21.5	22.1			23.4	23.7	24.4	24.7	24.8	24.8		24.8	24.8	24.8
Aid flows (in Billions of US dollars) 6/	0.4	0.4	0.4			0.3	0.3	0.2	0.2	0.1	0.2		0.2	0.3	
of which: Grants	0.2	0.2	0.1			0.2	0.2	0.1	0.1	0.1	0.1		0.1	0.2	
of which: Concessional loans	0.3	0.2	0.2			0.1	0.2	0.1	0.1	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 7/	...	...	...			1.5	1.1	0.8	0.8	0.7	0.8		0.6	0.6	0.6
Grant-equivalent financing (in percent of external financing) 7/	...	...	...			33.5	23.9	25.0	29.7	28.2	30.7		19.7	23.5	22.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	17.6	18.5	18.5			19.5	20.2	20.8	21.4	22.2	22.9		27.9	47.1	
Nominal dollar GDP growth	12.2	4.9	-0.1			5.2	3.7	2.9	3.3	3.4	3.4	3.6	5.1	5.4	4.9
PV of PPG external debt (in Billions of US dollars)			4.2			4.7	5.2	5.6	5.8	6.0	6.1		7.1	9.2	
(PVt-PVt-1)/GDPt-1 (in percent)						2.7	3.0	1.8	1.1	0.8	0.4	1.6	0.7	0.5	0.7
Gross workers' remittances (Billions of US dollars)	2.8	2.9	3.1			3.3	3.5	3.6	3.8	4.1	4.3		5.3	8.9	
PV of PPG external debt (in percent of GDP + remittances)	...	...	19.5			20.9	22.7	23.6	23.7	23.5	23.0		21.6	16.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	35.3			37.9	41.0	42.2	42.1	41.3	40.3		37.7	28.9	
Debt service of PPG external debt (in percent of exports + remittan	...	...	1.2			2.4	2.6	2.8	3.3	3.6	3.8		4.3	3.5	

Sources: National authorities and Fund staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Defined as grants, concessional loans, and debt relief.

7/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

8/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A3. Honduras: Sensitivity Analysis for Key Indicators of Public Debt 2014–34

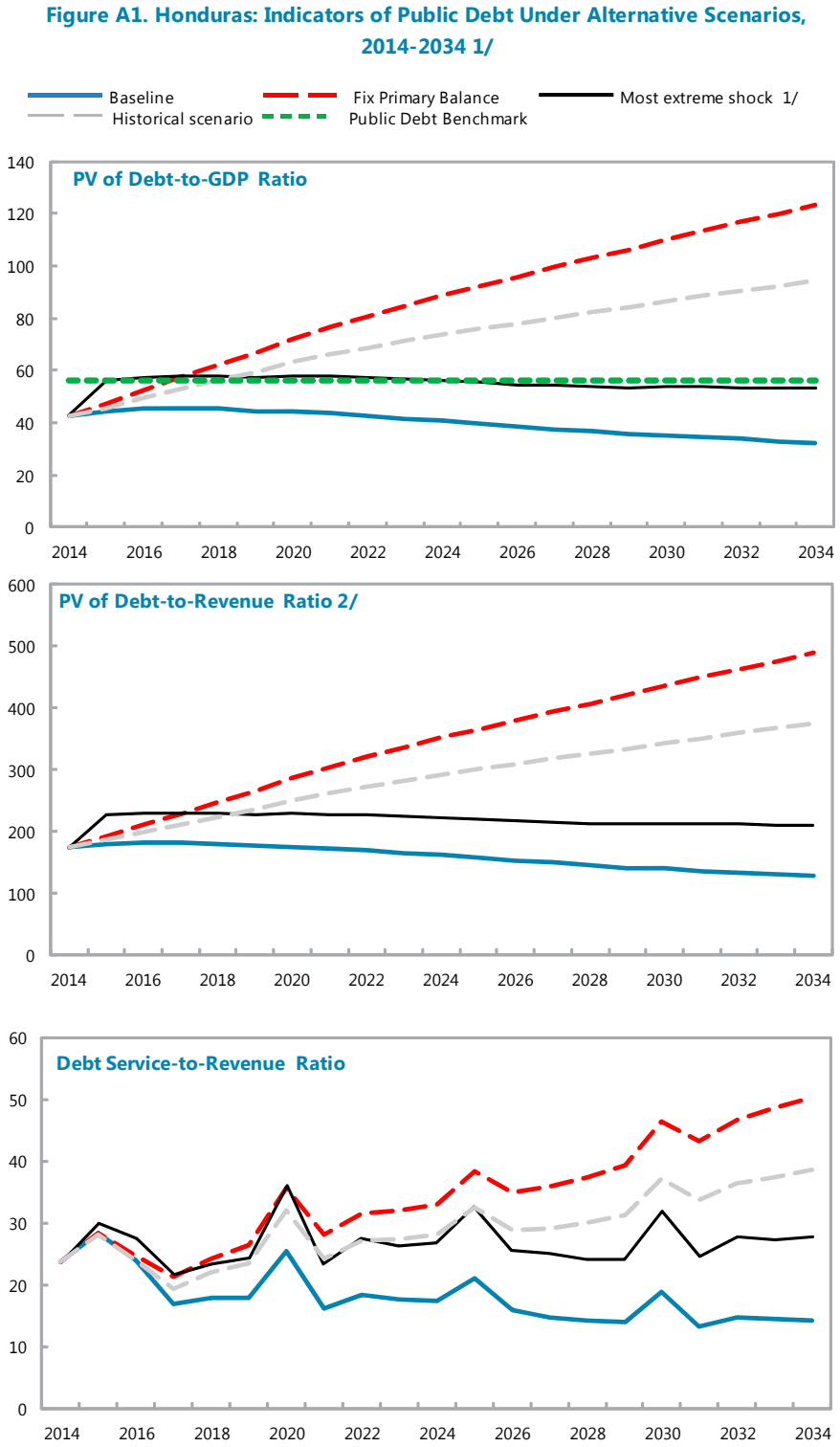
	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	43	44	45	46	45	45	41	32
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	43	46	49	53	56	59	74	94
A2. Primary balance is unchanged from 2014	43	47	53	58	62	67	89	123
A3. Permanently lower GDP growth 1/	43	45	46	47	48	48	51	67
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	43	45	48	49	50	50	50	49
B2. Primary balance is at historical average minus one standard deviations in 2015-201	43	48	53	54	54	53	50	41
B3. Combination of B1-B2 using one half standard deviation shocks	43	47	52	53	53	52	50	43
B4. One-time 30 percent real depreciation in 2015	43	56	57	58	58	57	56	53
B5. 10 percent of GDP increase in other debt-creating flows in 2015	43	54	55	56	56	55	52	43
PV of Debt-to-Revenue Ratio 2/								
Baseline	174	181	182	181	179	177	162	128
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	174	186	198	210	223	236	293	374
A2. Primary balance is unchanged from 2014	174	193	211	229	247	266	352	489
A3. Permanently lower GDP growth 1/	174	182	185	188	189	191	203	264
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	174	185	192	195	197	198	199	195
B2. Primary balance is at historical average minus one standard deviations in 2015-201	174	194	214	214	212	210	197	162
B3. Combination of B1-B2 using one half standard deviation shocks	174	192	209	210	209	207	197	168
B4. One-time 30 percent real depreciation in 2015	174	228	229	229	228	228	223	211
B5. 10 percent of GDP increase in other debt-creating flows in 2015	174	221	222	221	220	218	205	169
Debt Service-to-Revenue Ratio 2/								
Baseline	24	28	24	17	18	18	17	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	28	24	19	22	23	28	39
A2. Primary balance is unchanged from 2014	24	28	25	21	24	26	33	50
A3. Permanently lower GDP growth 1/	24	28	24	17	19	19	21	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	24	29	25	18	20	20	21	21
B2. Primary balance is at historical average minus one standard deviations in 2015-201	24	28	25	22	24	21	19	18
B3. Combination of B1-B2 using one half standard deviation shocks	24	28	25	21	23	20	20	19
B4. One-time 30 percent real depreciation in 2015	24	30	28	22	23	24	27	28
B5. 10 percent of GDP increase in other debt-creating flows in 2015	24	28	26	28	20	22	20	19

Sources: National authorities and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Figure A1. Honduras: Indicators of Public Debt Under Alternative Scenarios, 2014–34 1/



Sources: National authorities and Fund staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table A4. Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to-GDP+remittances ratio								
Baseline	21	23	24	24	23	23	**22**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	21	22	22	22	22	22	**24**	26
A2. New public sector loans on less favorable terms in 2014-2034 2	21	23	24	25	25	25	**26**	26
A3. Customized scenario /7	21	24	26	28	29	29	**31**	25
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	21	22	24	24	24	23	**22**	17
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	21	26	33	34	34	33	**32**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	21	22	23	23	23	22	**21**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	21	24	27	27	27	26	**25**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	21	23	25	25	25	24	**23**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	21	30	31	31	31	30	**28**	22
PV of debt-to-exports+remittances ratio								
Baseline	38	41	42	42	41	40	**38**	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	38	40	40	40	40	40	**45**	50
A2. New public sector loans on less favorable terms in 2014-2034 2	38	41	43	44	44	44	**46**	46
A3. Customized scenario /7	38	43	47	49	51	51	**54**	43
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	38	40	41	41	40	39	**37**	29
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	38	50	68	68	67	66	**64**	41
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	38	40	41	41	40	39	**37**	29
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	38	45	50	48	47	46	**44**	31
B5. Combination of B1-B4 using one-half standard deviation shocks	38	43	48	48	47	46	**44**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	38	40	41	41	40	39	**37**	29
PV of debt-to-revenue ratio								
Baseline	105	112	114	113	112	110	**104**	79
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	105	108	106	104	103	102	**110**	116
A2. New public sector loans on less favorable terms in 2014-2034 2	105	113	117	119	120	120	**127**	126
A3. Customized scenario /7	105	118	127	134	138	141	**152**	121
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	105	111	115	114	113	111	**106**	81
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	105	127	161	161	160	159	**155**	100
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	105	110	109	109	108	106	**101**	77
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	105	119	129	129	128	126	**122**	86
B5. Combination of B1-B4 using one-half standard deviation shocks	105	113	118	118	117	115	**111**	80
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	105	156	158	157	156	153	**146**	112

Table A4. Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (continued)

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
Debt service-to-exports+remittances ratio								
Baseline	2	3	3	3	4	4	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	2	3	3	3	3	3	**4**	4
A2. New public sector loans on less favorable terms in 2014-2034 2	2	3	3	3	3	3	**4**	5
A3. Customized scenario /7	2	3	3	4	4	4	**5**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	2	3	3	3	4	4	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	2	3	4	5	5	5	**6**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	2	3	3	3	4	4	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	2	3	3	4	4	4	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	3	4	4	4	**5**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	2	3	3	3	4	4	**4**	4
Debt service-to-revenue ratio								
Baseline	7	7	8	9	10	10	**12**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	7	7	7	8	8	9	**9**	10
A2. New public sector loans on less favorable terms in 2014-2034 2	7	7	7	8	8	9	**12**	13
A3. Customized scenario /7	7	7	8	10	11	12	**15**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	7	7	8	9	10	11	**12**	10
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	7	7	8	11	12	13	**15**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	7	7	7	9	10	10	**12**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	7	7	8	10	11	11	**13**	11
B5. Combination of B1-B4 using one-half standard deviation shocks	7	7	8	9	10	11	**12**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	7	10	11	13	14	15	**17**	14
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	3	3	3	3	3	3	**3**	3

Sources: National authorities and Fund staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

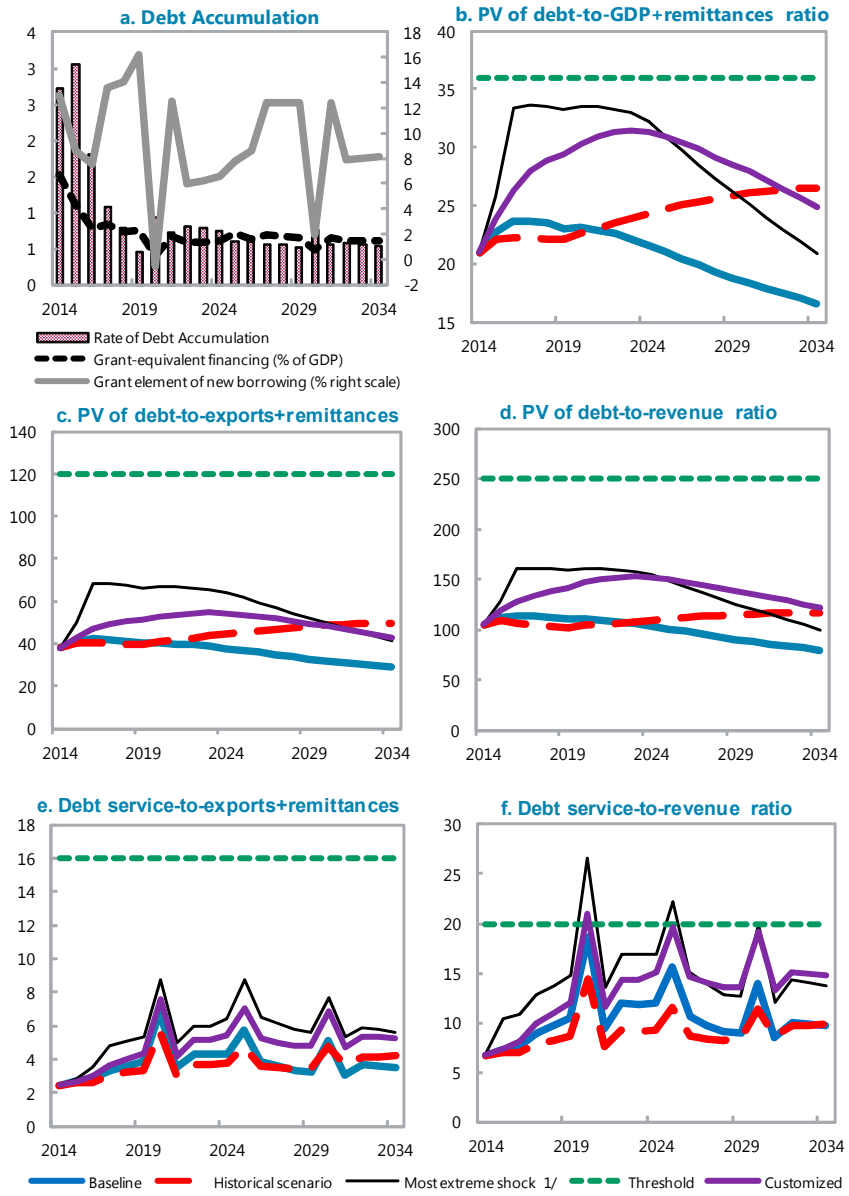
4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

7/ In the customized scenario, (i) real GDP growth is 2.5 percent from 2014 to 2017, and rises 0.2 percentage points per year to reach 3.3 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, with the rate of interest on Eurobonds rising to 8.5 percent, starting in 2015; (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the baseline (this increase fades out partially at a constant rate of 15 percent during the DSA period, e.g., compared to the baseline, the deficit is higher by 1.7, 1.0 and 0.5 percent of GDP respectively in 2016, 2019 and 2024; and (iv) contingent liabilities (estimated at about 2 percent of GDP) related to existing PPP projects are assumed to fully materialize over the next three years and result in an increase in public external debt.

Figure A2. Honduras: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014–34 1/



Sources: Country authorities and Fund staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock
2/ For the customized scenario see footnote 7/ in table A4.